

Exercise 1 — Basic Class

```
ASSIGNMENT5 >  Q1.cpp >  main()
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      string name;
7      int age;
8      string course;
9  };
10
11 int main() {
12     Student s1;
13
14     s1.name = "Simmi";
15     s1.age = 21;
16     s1.course = "B.Tech CSE";
17
18     cout << "Name: " << s1.name << endl;
19     cout << "Age: " << s1.age << endl;
20     cout << "Course: " << s1.course << endl;
21
22     return 0;
23 }
```

```
● mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q1
Name: Simmi
Age: 21
Course: B.Tech CSE
```

Exercise 2 — Multiple Objects

```

ASSIGNMENT5 > G+ Q2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      string name;
7      int age;
8      string course;
9  };
10
11 int main() {
12     Student s1, s2, s3;
13
14     s1.name = "Simmi";
15     s1.age = 21;
16     s1.course = "B.Tech CSE";
17
18     s2.name = "Rahul";
19     s2.age = 20;
20     s2.course = "B.Tech IT";
21
22     s3.name = "Anjali";
23     s3.age = 22;
24     s3.course = "BCA";
25
26     cout << "Student 1" << endl;
27     cout << "Name: " << s1.name << endl;
28     cout << "Age: " << s1.age << endl;
29     cout << "Course: " << s1.course << endl;
30
31     cout << "\nStudent 2" << endl;
32     cout << "Name: " << s2.name << endl;
33     cout << "Age: " << s2.age << endl;
34     cout << "Course: " << s2.course << endl;
35
36     cout << "\nStudent 3" << endl;
37     cout << "Name: " << s3.name << endl;
38     cout << "Age: " << s3.age << endl;
39     cout << "Course: " << s3.course << endl;
40
41     return 0;
42 }

```

```

● mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q2
Student 1
Name: Simmi
Age: 21
Course: B.Tech CSE

Student 2
Name: Rahul
Age: 20
Course: B.Tech IT

Student 3
Name: Anjali
Age: 22
Course: BCA

```

Exercise 3 – Method

```

ASSIGNMENT5 > G+ Q3.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      string name;
7      int age;
8      string course;
9
10     void display() {
11         cout << "Name: " << name << endl;
12         cout << "Age: " << age << endl;
13         cout << "Course: " << course << endl;
14     }
15 };
16
17 int main() {
18     Student s1;
19
20     s1.name = "Simmi";
21     s1.age = 21;
22     s1.course = "B.Tech CSE";
23
24     s1.display();
25
26     return 0;
27 }

```

```

mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q3
Name: Simmi
Age: 21
Course: B.Tech CSE

```

Exercise 4 – Constructor

```

ASSIGNMENT5 > G+ Q4.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      string name;
7      int age;
8      string course;
9
10     // Constructor
11     Student(string name, int age, string course) {
12         this->name = name;
13         this->age = age;
14         this->course = course;
15     }
16
17     void display() {
18         cout << "Name: " << name << endl;
19         cout << "Age: " << age << endl;
20         cout << "Course: " << course << endl;
21     }
22 };
23
24 int main() {
25     // Object initialized using constructor
26     Student s1("Simmi", 21, "B.Tech CSE");
27
28     s1.display();
29
30     return 0;
31 }

```

```
mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q4
Name: Simmi
Age: 21
Course: B.Tech CSE
```

Exercise 5 – Employee

```
ASSIGNMENT5 > Q5.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class Employee {
5  public:
6      int id;
7      string name;
8      double salary;
9      string department;
10
11     // Parameterized Constructor
12     Employee(int id, string name, double salary, string department) {
13         this->id = id;
14         this->name = name;
15         this->salary = salary;
16         this->department = department;
17     }
18
19     // Display Method
20     void display() {
21         cout << "Employee ID: " << id << endl;
22         cout << "Name: " << name << endl;
23         cout << "Salary: " << salary << endl;
24         cout << "Department: " << department << endl;
25     }
26 };
27
28 int main() {
29     Employee e1(101, "Simmi", 45000, "Computer Science");
30
31     e1.display();
32
33     return 0;
34 }
```

```
mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q5
Employee ID: 101
Name: Simmi
Salary: 45000
Department: Computer Science
mac@SimmiTiwari ASSIGNMENT5 %
```

Exercise 6 – BankAccount

ASSIGNMENT5 > Q6.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  class BankAccount {
5  public:
6      string accountNumber;
7      string accountHolder;
8      double balance;
9
10     // Parameterized Constructor
11     BankAccount(string accountNumber, string accountHolder, double balance) {
12         this->accountNumber = accountNumber;
13         this->accountHolder = accountHolder;
14         this->balance = balance;
15     }
16
17     // Display Method
18     void display() {
19         cout << "Account Number: " << accountNumber << endl;
20         cout << "Account Holder: " << accountHolder << endl;
21         cout << "Balance: " << balance << endl;
22     }
23
24     // Deposit Method
25     void deposit(double amount) {
26         balance += amount;
27         cout << "Deposited: " << amount << endl;
28     }
29
30     // Withdraw Method
31     void withdraw(double amount) {
32         if (amount <= balance) {
33             balance -= amount;
34             cout << "Withdrawn: " << amount << endl;
35         } else {
36             cout << "Insufficient balance!" << endl;
37         }
38     }
39 };
```

```
41 int main() {
42     BankAccount account("123456789", "Simmi", 10000);
43
44     account.display();
45
46     cout << "\nAfter Deposit:" << endl;
47     account.deposit(2000);
48     account.display();
49
50     cout << "\nAfter Withdrawal:" << endl;
51     account.withdraw(3000);
52     account.display();
53
54     return 0;
55 }
```

```
mac@SimmiTiwari ASSIGNMENT5 % /Users/mac/c++_college/ASSIGNMENT5/Q6
Account Number: 123456789
Account Holder: Simmi
Balance: 10000

After Deposit:
Deposited: 2000
Account Number: 123456789
Account Holder: Simmi
Balance: 12000

After Withdrawal:
Withdrawn: 3000
Account Number: 123456789
Account Holder: Simmi
Balance: 9000
```